Lecture 16 (10/21)

Leaving out predictors and the sample distribution

Last class:

- Omitted variables in AR processes: bitcoin, Ukraine and Gaza
- Incumbency advantage in congressional elections; categorical predictors and dummy variables

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Today:

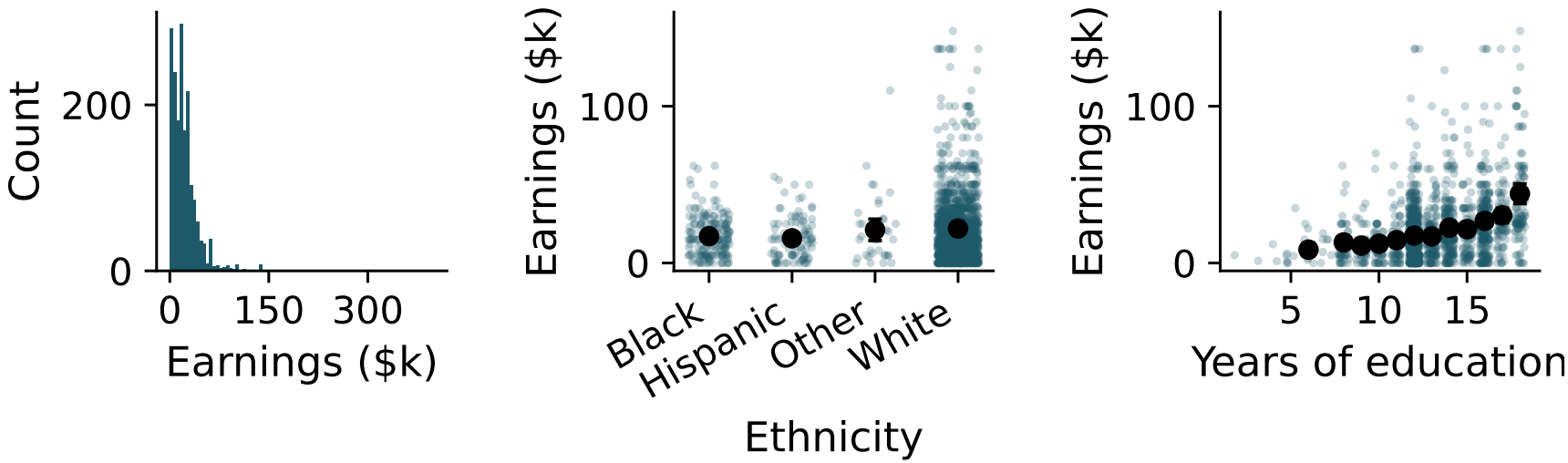
1) Leaving out all predictors but one: earnings, elections, test scores
2) The sample distribution with correlated predictors

Example

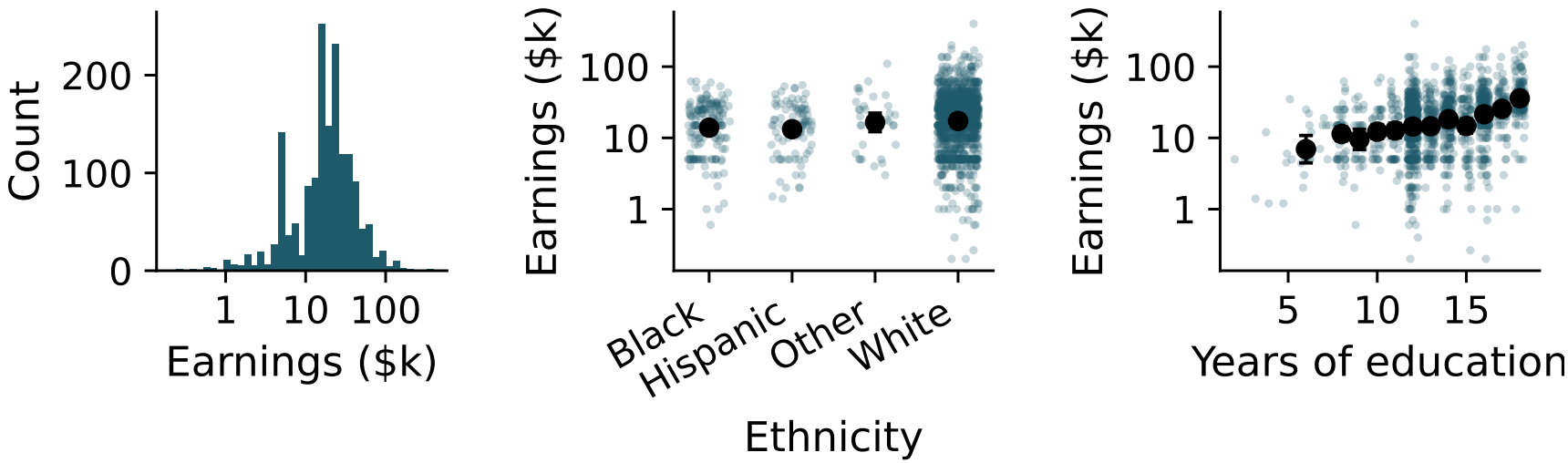
How are earnings related to gender and education, and what else could explain these associations? Survey of 1814 US adults (from *Regression and Other Stories*):

- Y (`earnk`): earnings in \$1000s;
- $X_{\text{hispanic}}, X_{\text{other}}, X_{\text{white}}$ (`hispanic`, `other`, `white`): dummy variables for ethnicity, with Black as the baseline;
- X_{male} (`male`), X_{height} (`height`, inches), X_{educ} (`education`, years), X_{age} (`age`, years).

Earnings vs. ethnicity and education



Linear scale. Note: 4 respondents earning more than $160k are not shown.



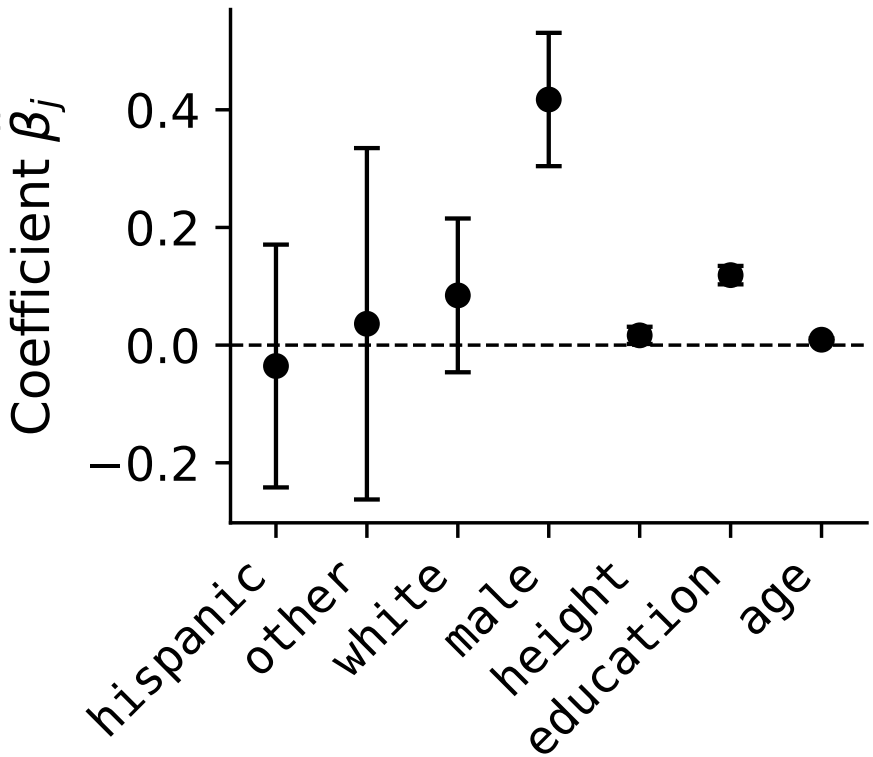
Log scale using 1627 respondents with positive earnings. Black dots: averages with 95% CIs.

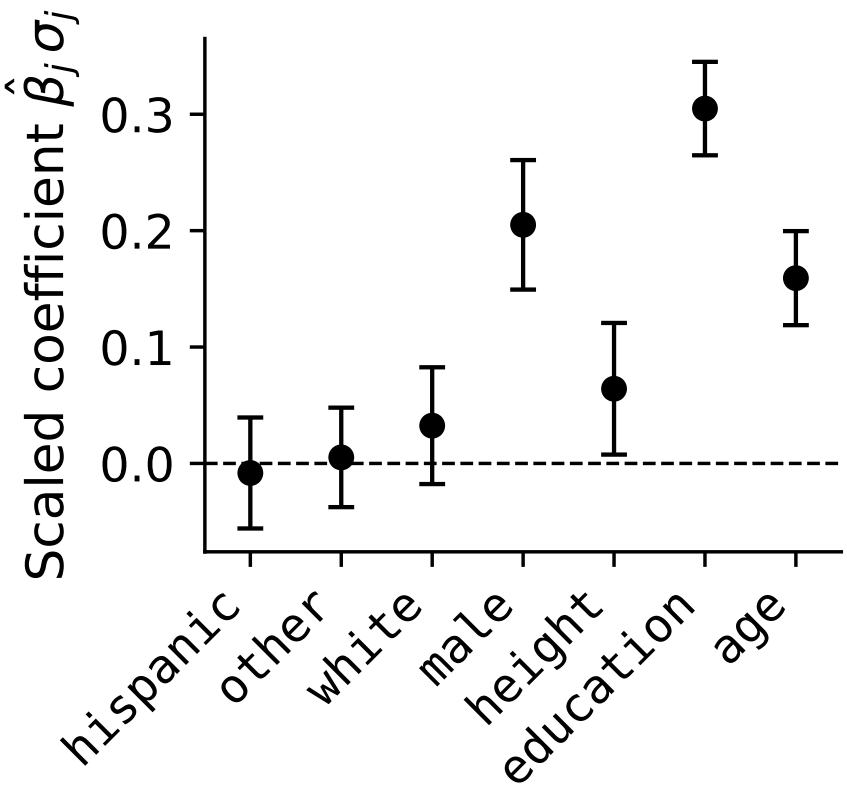
Earnings with all predictors

$$\log(\texttt{earnk}) = \beta_0 + \beta_{\text{hispanic}} X_{\text{hispanic}} + \beta_{\text{other}} X_{\text{other}} + \beta_{\text{white}} X_{\text{white}} + \beta_{\text{male}} X_{\text{male}} + \beta_{\text{height}} X_{\text{height}} + \beta_{\text{educ}} X_{\text{educ}} + \beta_{\text{age}} X_{\text{age}} + \epsilon$$

Dep. Variable:		log_earnk	R-squared:			0.211
	coef	std err	t	P>|t|	[0.025	0.975]
const	-0.5156	0.496	-1.039	0.299	-1.489	0.458
hispanic	-0.0355	0.105	-0.337	0.736	-0.242	0.171
other	0.0363	0.152	0.238	0.812	-0.263	0.335
white	0.0845	0.067	1.268	0.205	-0.046	0.215
male	0.4175	0.058	7.222	0.000	0.304	0.531
height	0.0166	0.007	2.223	0.026	0.002	0.031
education	0.1189	0.008	14.885	0.000	0.103	0.135
age	0.0092	0.001	7.724	0.000	0.007	0.012

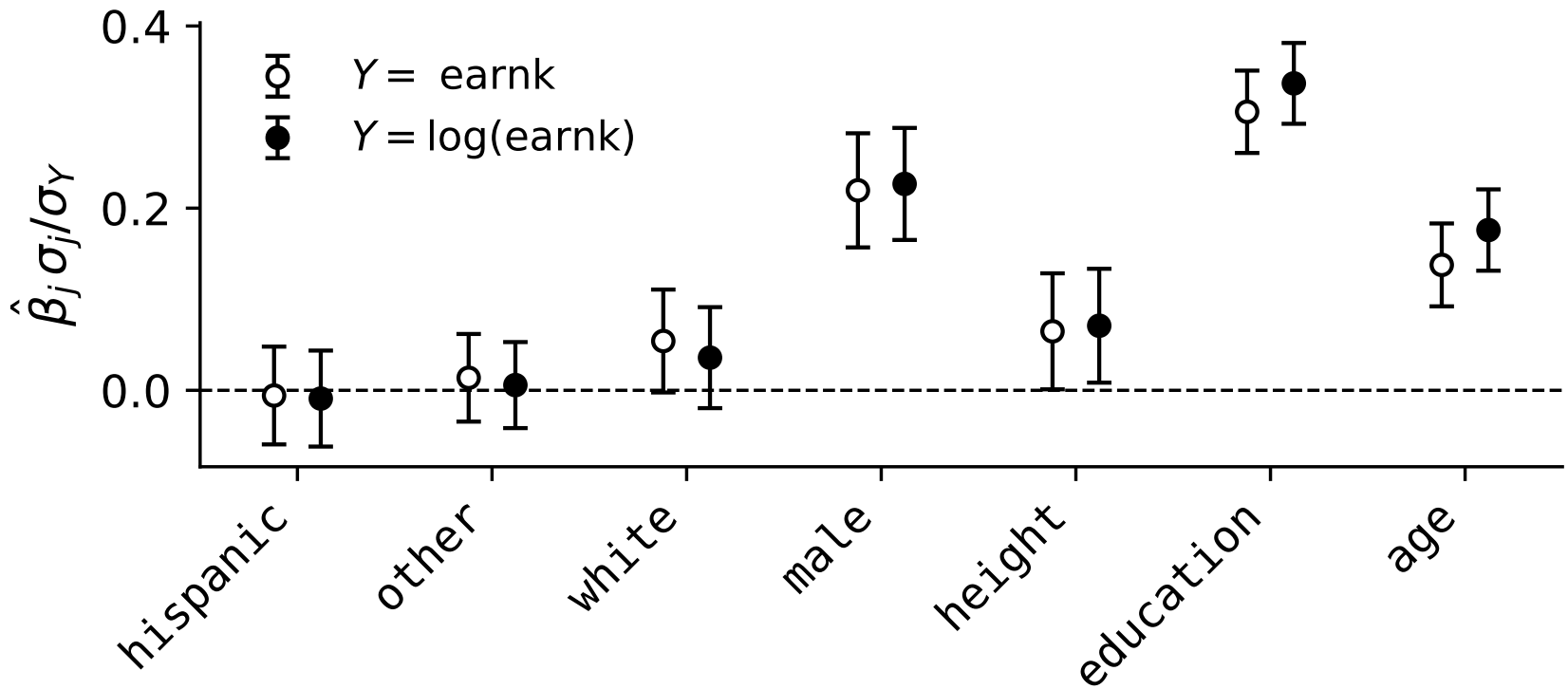
Coefficients vs. scaled coefficients





Standardized coefficients: earnk vs. log(earnk)

The full model fit to the 1627 respondents with positive earnings, with $Y =$ earnk ($R^2 = 0.18$) and with $Y =$ log(earnk) ($R^2 = 0.21$). Each coefficient is scaled by σ_j/σ_Y:



Omitting many predictors

Theorem

Suppose $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_K X_K + \epsilon$, *where* ϵ *is uncorrelated with* $X_1, \ldots, X_K$.

Omitted variable bias: if $X_2, \ldots, X_K$ *are left out, the slope of the regression of* Y *on* X_1 *is*

$$\boxed{\beta_1' = \beta_1 + \sum_{j=2}^{K} \beta_j \beta_{1,j}'}, \qquad \beta_{1,j}' = \frac{cov(X_1, X_j)}{var(X_1)}.$$

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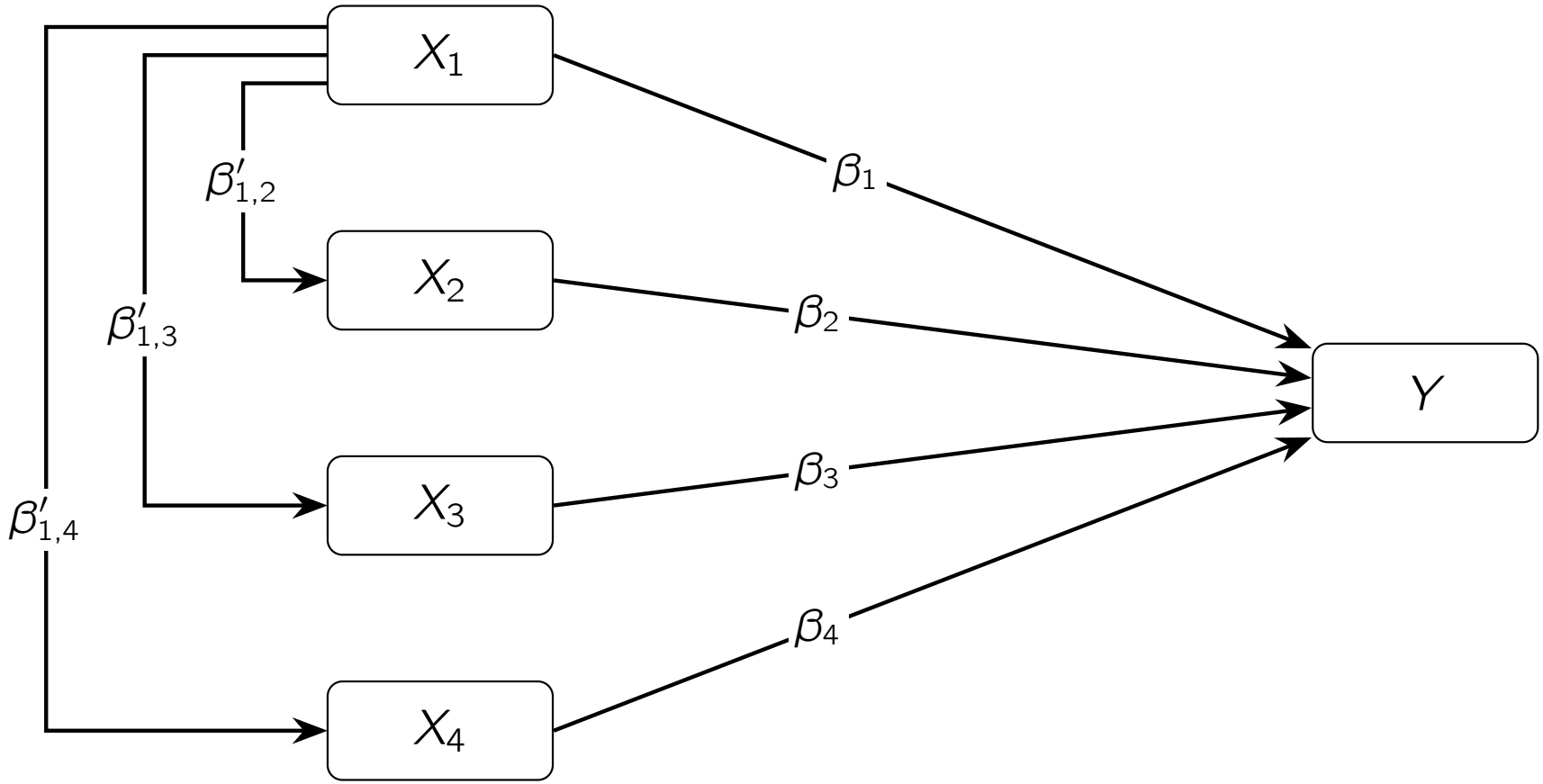
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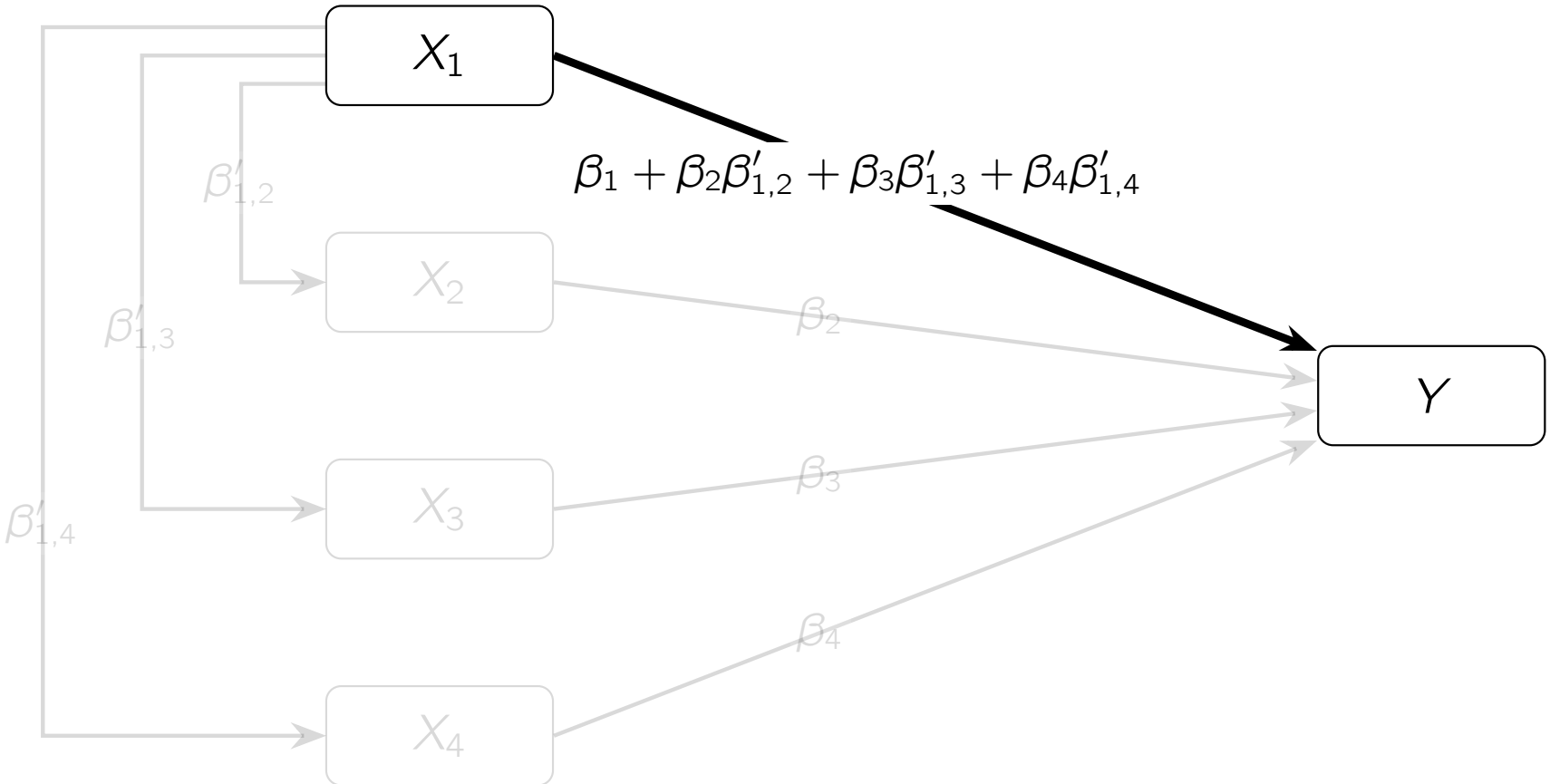
Question

Derive this formula, assuming $E[X_j] = 0$ for $j = 1, \ldots, K$ and $\beta_0 = 0$ (hence $E[Y] = 0$).

Leaving out all predictors but one

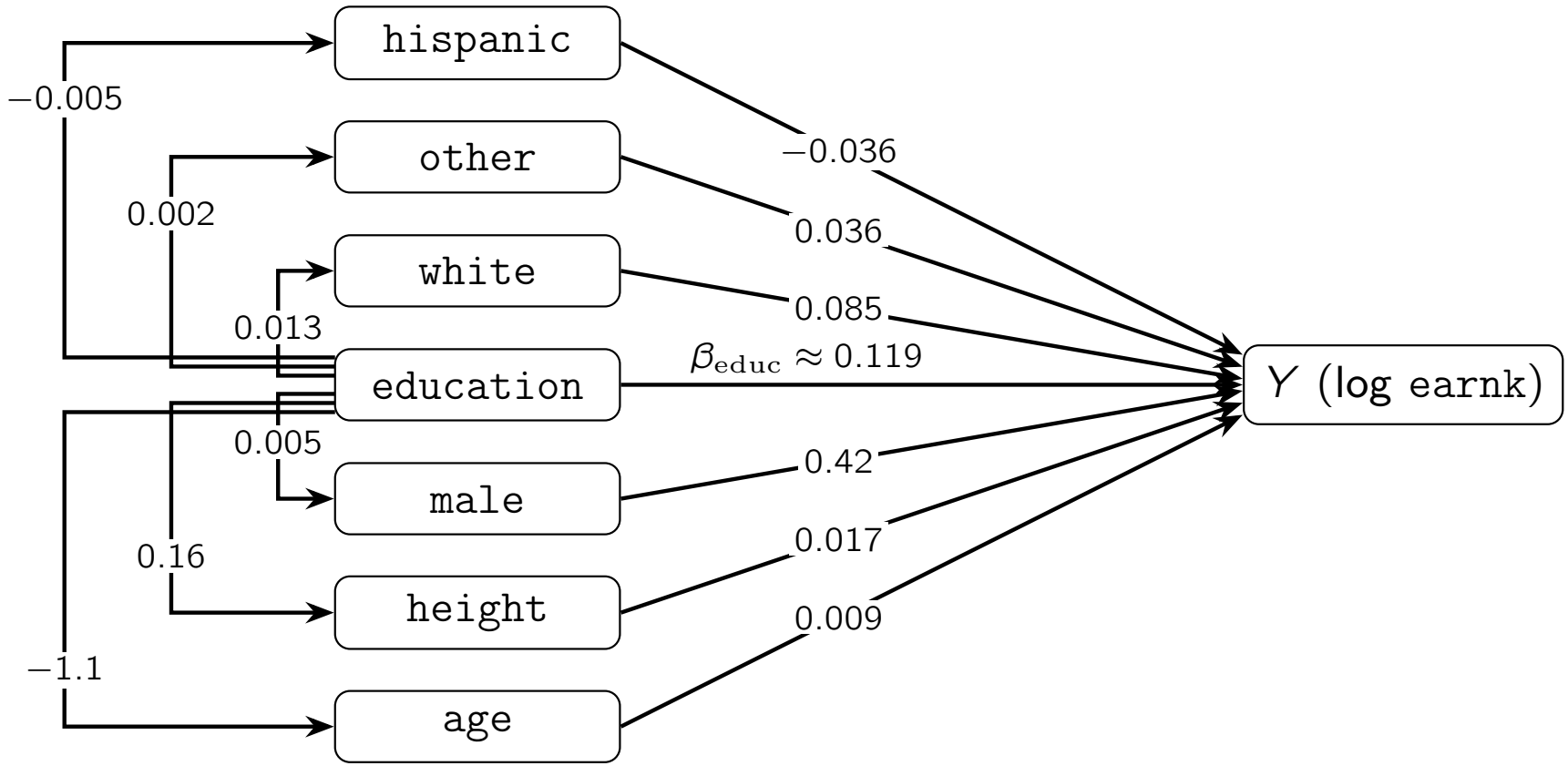


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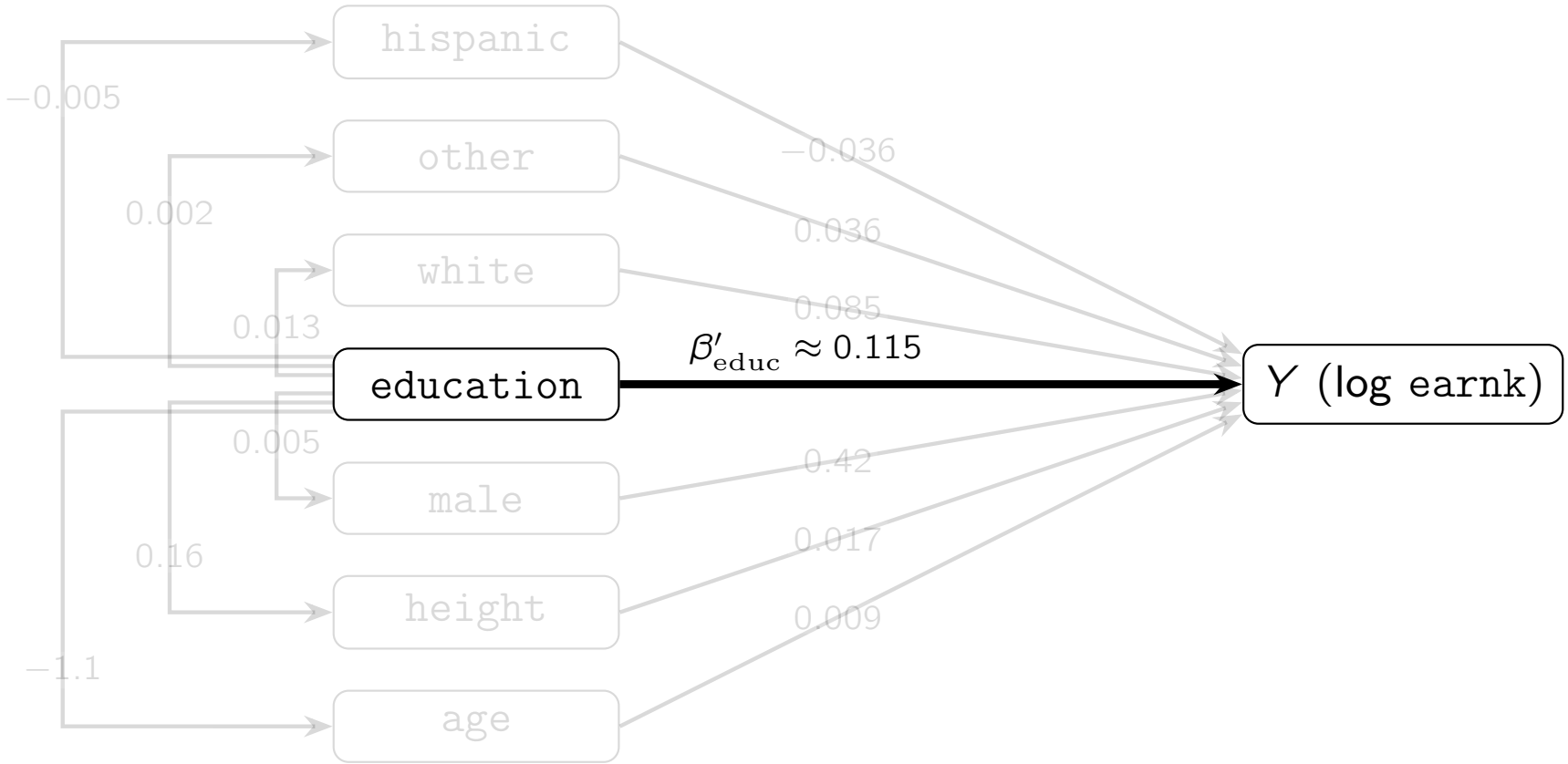
Here $\beta'_{1,j}$ is the single-predictor slope of X_j on X_1: this is the theorem for K predictors.

Leaving out all predictors but education



Left arrows: single-predictor slopes $\beta'_{\text{educ},j}$ of each predictor on `education`. Right arrows: coefficients β_j in the full model.

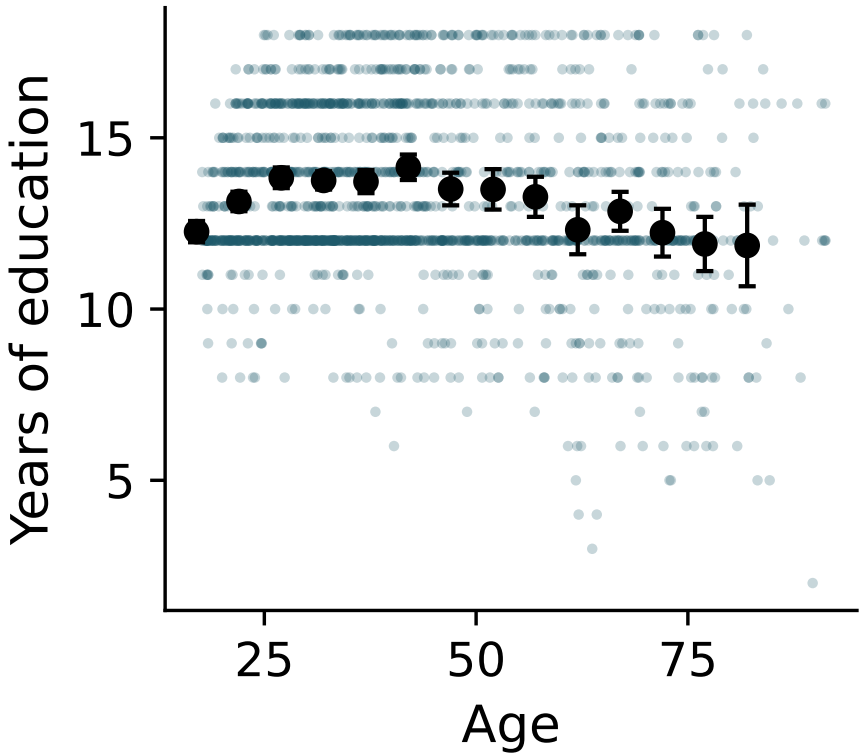
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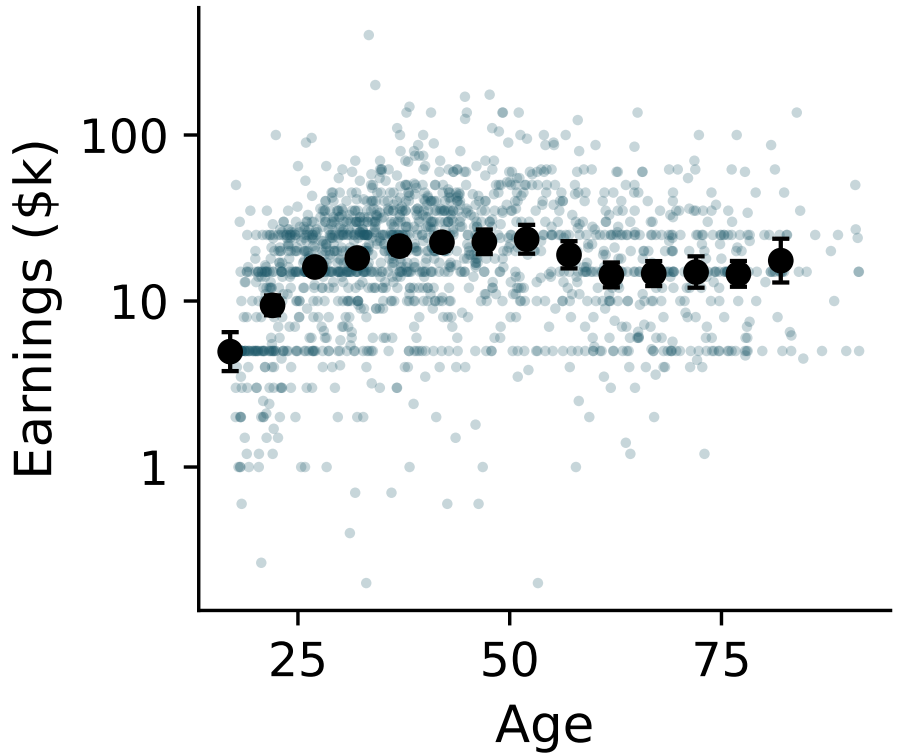


Left arrows: single-predictor slopes $\beta'_{\text{educ},j}$ of each predictor on `education`. Right arrows: coefficients β_j in the full model.

$$\underbrace{0.115}_{\beta'_{\text{educ}}} \approx \underbrace{0.119}_{\beta_{\text{educ}}} + \underbrace{0.001}_{\text{ethnicity (3 dummies)}} + \underbrace{0.42 \times 0.005}_{\text{male} \approx 0.002} + \underbrace{0.017 \times 0.16}_{\text{height} \approx 0.003} + \underbrace{0.009 \times (-1.1)}_{\text{age} \approx -0.010}$$

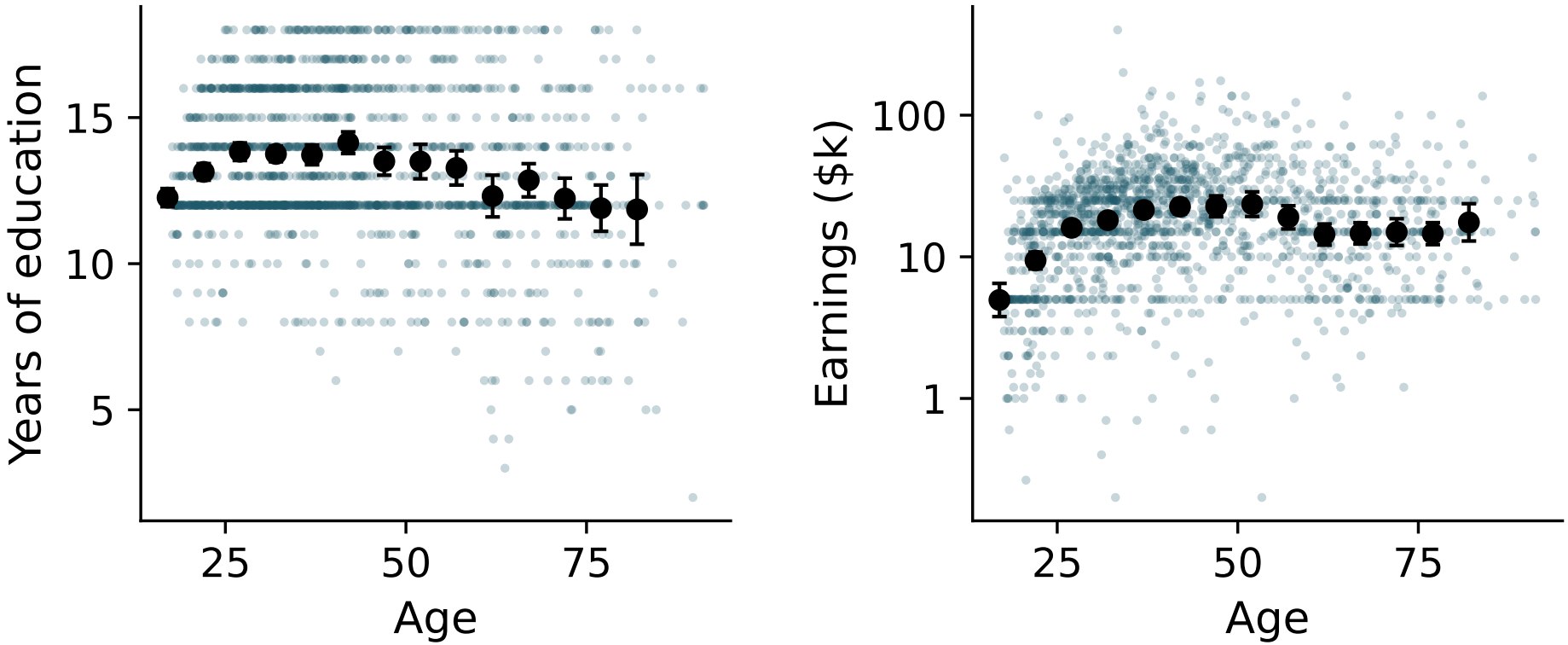
Education and earnings vs. age





Respondents with positive earnings; earnings on a log scale. Black dots: 5-year averages with 95% CIs.

Education and earnings vs. age

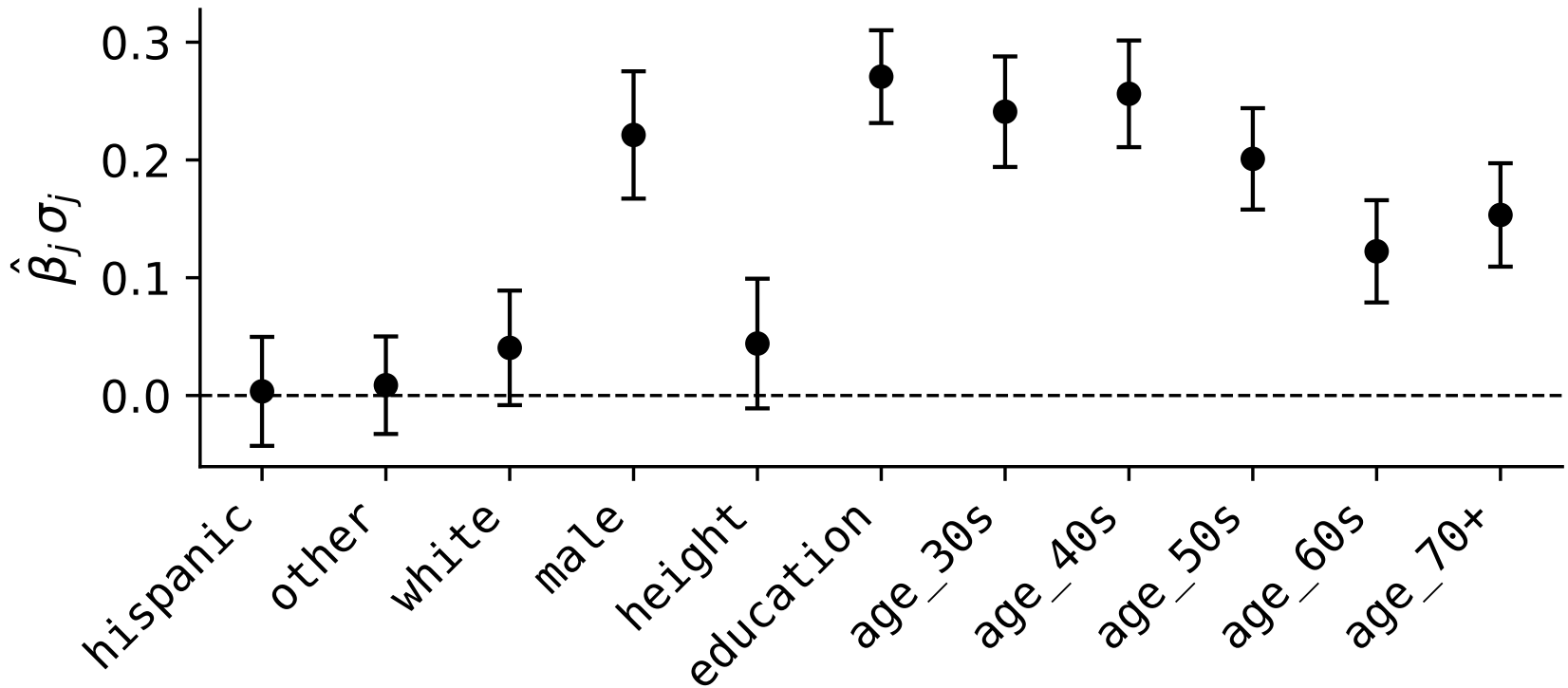


Respondents with positive earnings; earnings on a log scale. Black dots: 5-year averages with 95% CIs.

Explanation: Older respondents are from earlier generations, when less schooling was typical (the youngest are partly still in school).

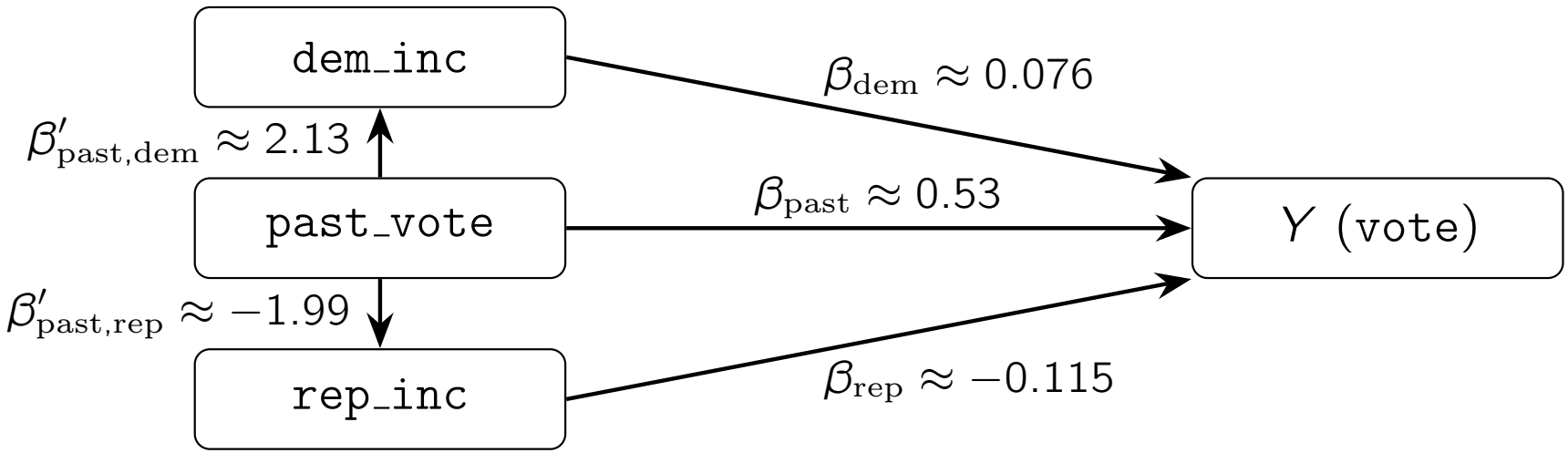
Age as a categorical predictor

Replace age by one dummy variable per decade (baseline: 18–29, with 70 and over pooled), keeping the other predictors:



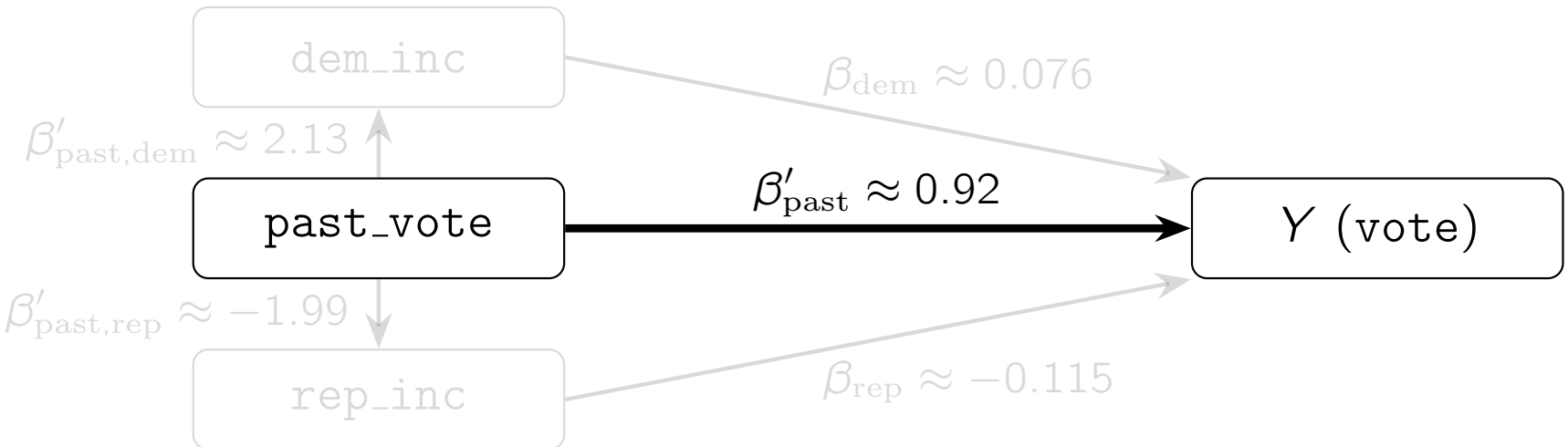
Revisiting elections: comparison to past vote only

Dep. Variable:		vote	R-squared:		0.834
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0382	0.012	3.311	0.001	0.016	0.061
past_vote	0.9182	0.020	46.584	0.000	0.879	0.957



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$$\underbrace{0.92}_{\beta'_{\text{past}}} \approx \underbrace{0.53}_{\beta_{\text{past}}} + \underbrace{0.076 \times 2.13}_{\beta_{\text{dem}}\beta'_{\text{past,dem}} \approx 0.16} + \underbrace{(-0.115) \times (-1.99)}_{\beta_{\text{rep}}\beta'_{\text{past,rep}} \approx 0.23}$$

Districts with a high past vote tend to have Democratic incumbents, and incumbency raises the vote.

Revisiting children's test scores: mother age

Children's test scores again, now adding X_{age} (mom_age), the mother's age at birth.

Dep. Variable:		kid_score	R-squared:			0.008
	coef	std err	t	P>|t|	[0.025	0.975]
const	70.9569	8.307	8.542	0.000	54.631	87.283
mom_age	0.6952	0.362	1.920	0.055	-0.016	1.407

Dep. Variable:		kid_score	R-squared:			0.215
	coef	std err	t	P>|t|	[0.025	0.975]
const	20.9847	9.130	2.298	0.022	3.039	38.930
mom_hs	5.6472	2.258	2.501	0.013	1.210	10.085
mom_iq	0.5625	0.061	9.276	0.000	0.443	0.682
mom_age	0.2248	0.331	0.680	0.497	-0.425	0.875

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Question

Interpret the coefficient on `mom_age` in each model. With only `mom_hs` and `mom_iq`, the coefficients were 5.95 and 0.564 and $R^2 = 0.214$. Why did adding `mom_age` change so little even though age is associated with test scores in the first regression? What does this tell us about association between age and the other predictors?

Mother's age vs. schooling and IQ

Dep. Variable:		mom_hs	R-squared:			0.046
	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0423	0.164	0.258	0.797	-0.280	0.365
mom_age	0.0326	0.007	4.565	0.000	0.019	0.047
Dep. Variable:		mom_iq	R-squared:			0.008
	coef	std err	t	P>|t|	[0.025	0.975]
const	88.4081	6.105	14.482	0.000	76.409	100.407
mom_age	0.5087	0.266	1.912	0.057	-0.014	1.032

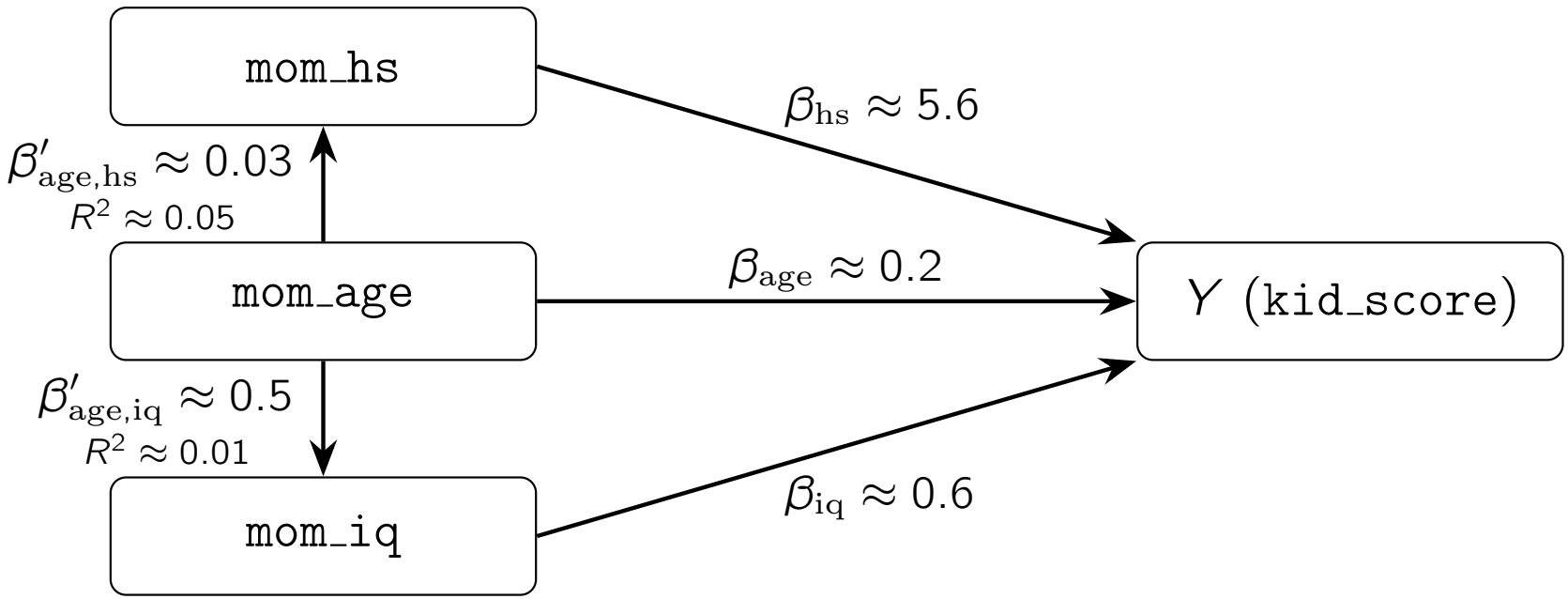
Mother's age vs. schooling and IQ

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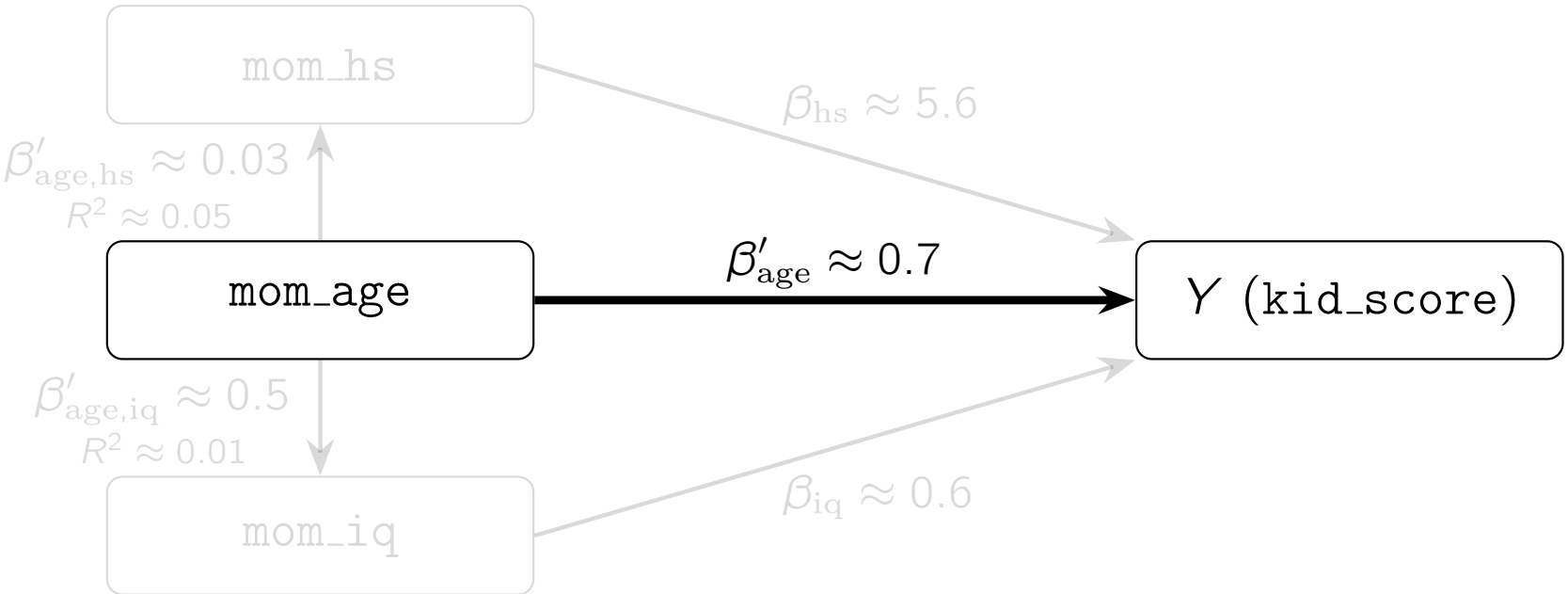
Question

Use the omitted variable bias formula to write the slope $\beta'_{\text{age}} \approx 0.70$ of the single-predictor regression of `kid_score` on `mom_age` in terms of the coefficients of the three-predictor model and the slopes $\beta'_{\text{age,hs}}$ and $\beta'_{\text{age,iq}}$ above.

How mother's age is related to test scores



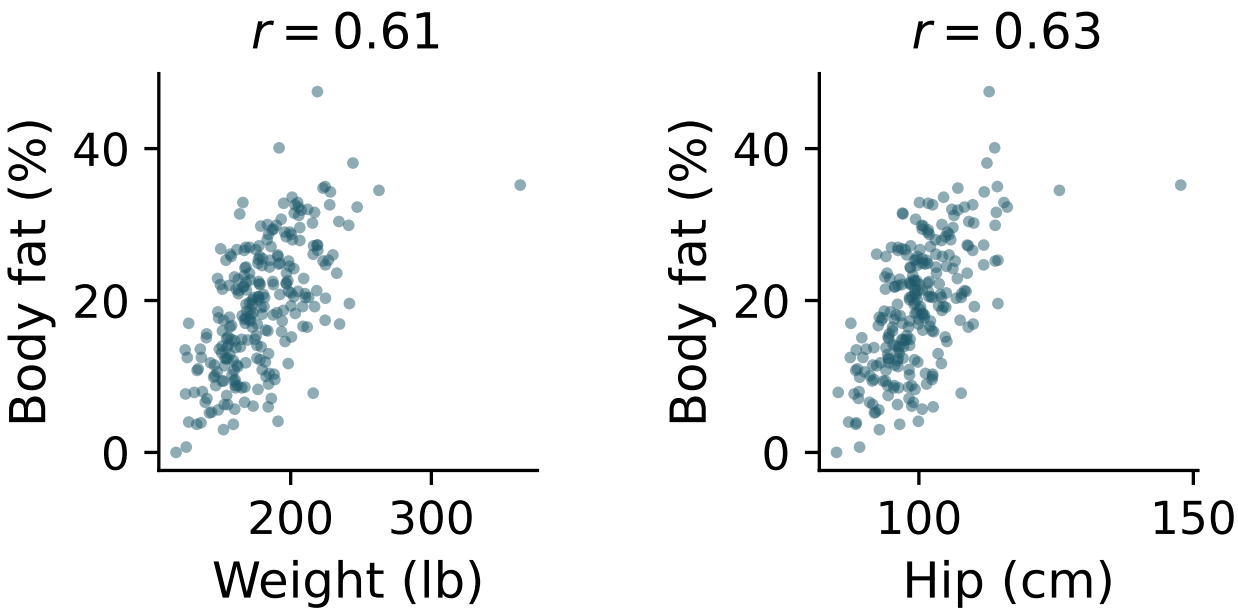
How mother's age is related to test scores



$$\underbrace{0.7}_{\beta'_{\text{age}}} \approx \underbrace{0.2}_{\beta_{\text{age}}} + \underbrace{0.03 \times 5.6}_{\beta'_{\text{age,hs}}\beta_{\text{hs}} \,\approx\, 0.2} + \underbrace{0.5 \times 0.6}_{\beta'_{\text{age,iq}}\beta_{\text{iq}} \,=\, 0.3}$$

Body fat, weight and hip size

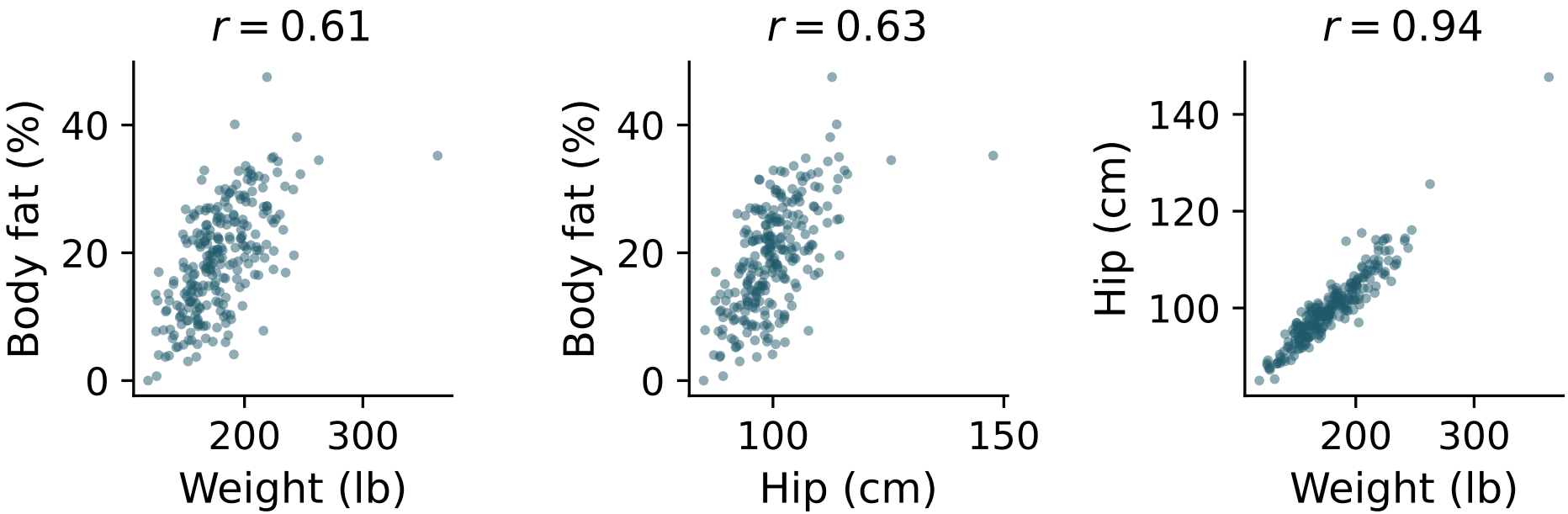
Percent body fat (from underwater weighing) of 252 men, with their weight (lb) and hip circumference (cm):



Data: Johnson (1996), *Journal of Statistics Education*.

Body fat, weight and hip size

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Body fat: one predictor at a time vs. both

Dep. Variable:		bodyfat	R-squared:			0.375
	coef	std err	t	P>|t|	[0.025	0.975]
const	-12.0516	2.581	-4.669	0.000	-17.136	-6.968
weight	0.1744	0.014	12.249	0.000	0.146	0.202

Dep. Variable:		bodyfat	R-squared:			0.391
	coef	std err	t	P>|t|	[0.025	0.975]
const	-53.8129	5.775	-9.318	0.000	-65.187	-42.438
hip	0.7303	0.058	12.666	0.000	0.617	0.844

Dep. Variable:		bodyfat	R-squared:			0.396
	coef	std err	t	P>|t|	[0.025	0.975]
const	-41.4134	10.317	-4.014	0.000	-61.734	-21.093
weight	0.0600	0.041	1.449	0.149	-0.022	0.142
hip	0.4988	0.170	2.936	0.004	0.164	0.833

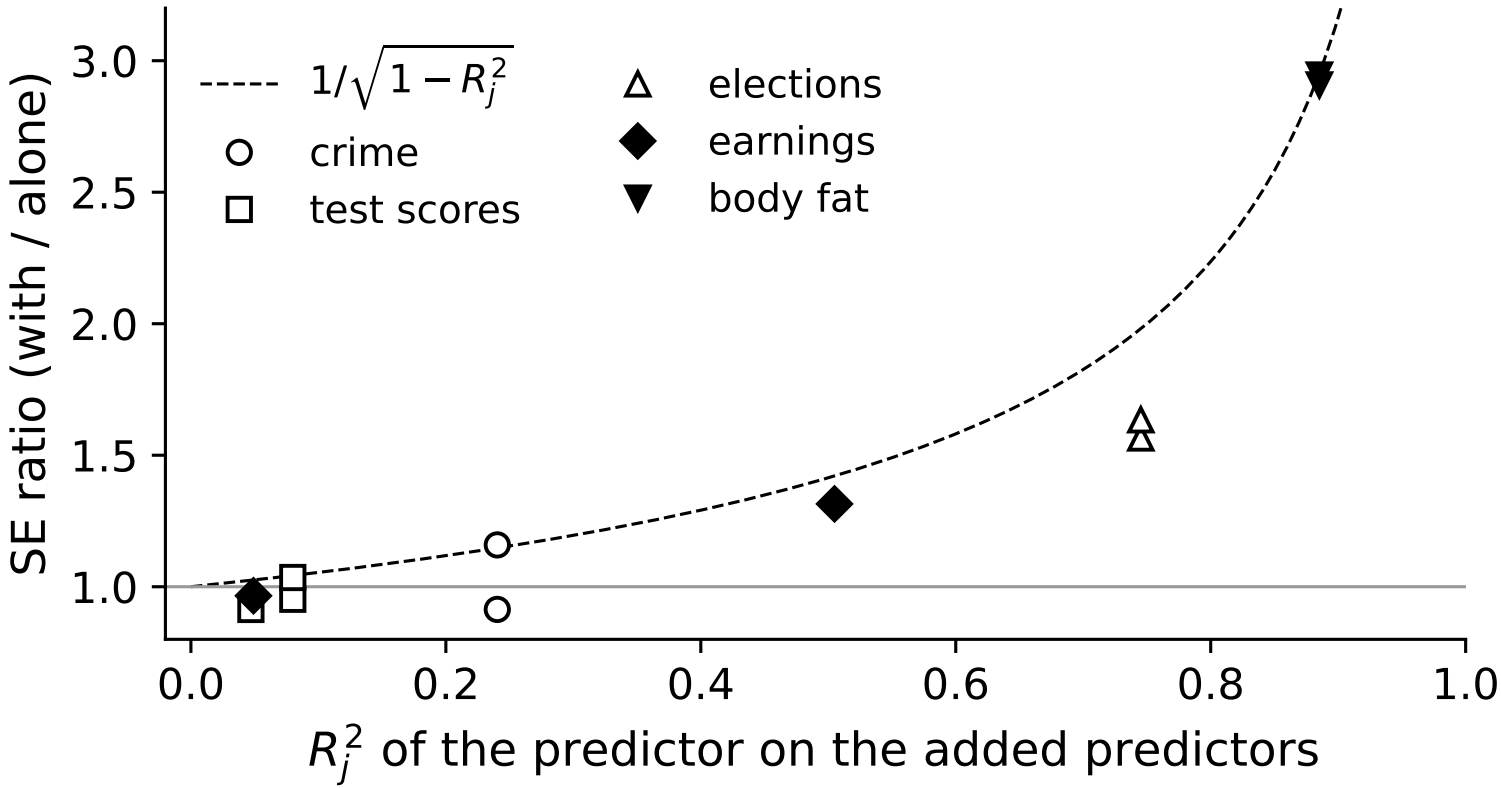
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Question

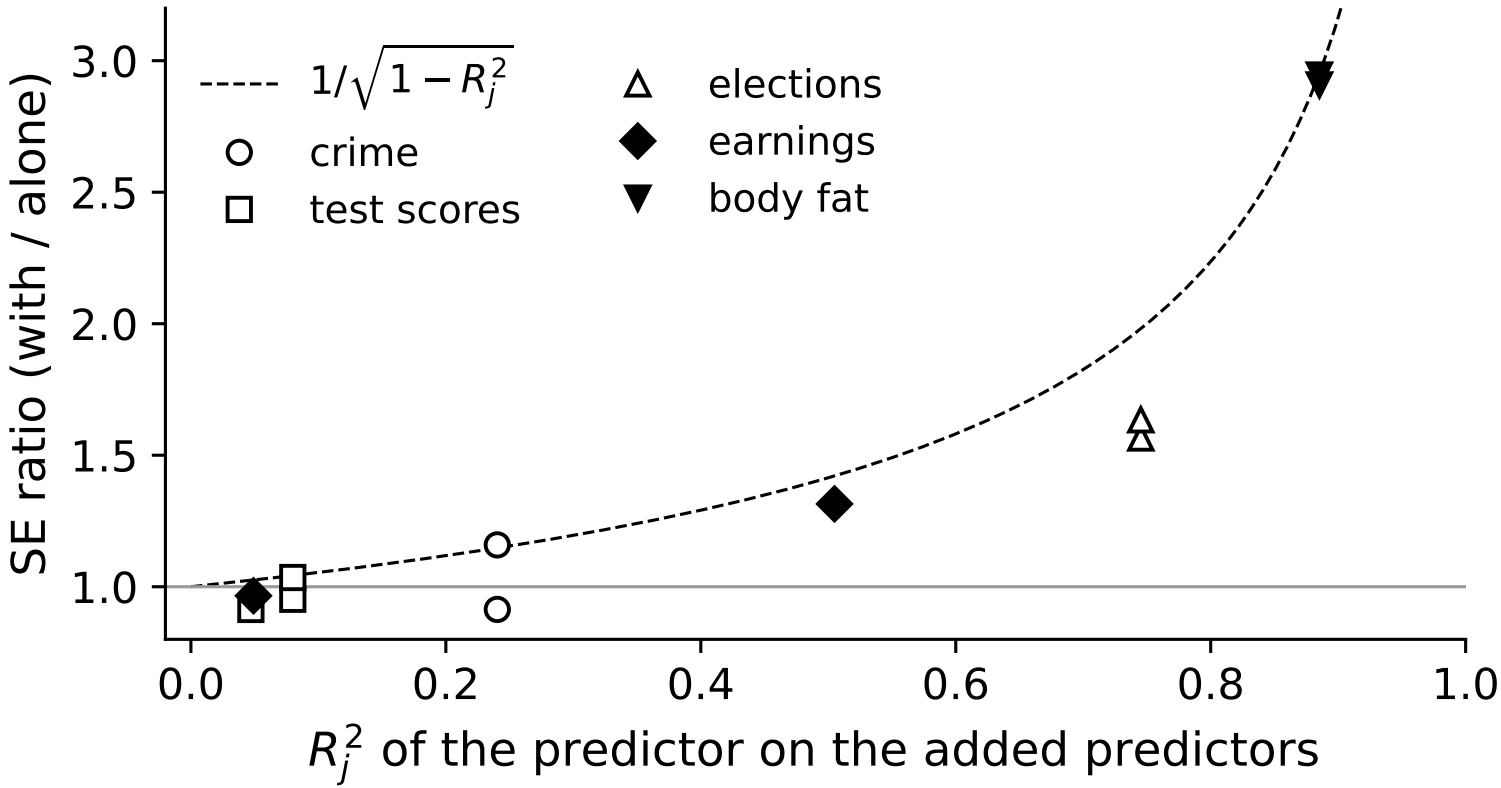
On its own, each predictor is strongly related to body fat. Why does the standard error of the weight coefficient triple, and weight stop being significant, when both are in the model?

Adding predictors and the standard error



Each point: one coefficient from our examples, its standard error with the other predictors divided by its standard error alone. R_j^2: from regressing that predictor on the added ones.

Adding predictors and the standard error



Each point: one coefficient from our examples, its standard error with the other predictors divided by its standard error alone. R_j^2: from regressing that predictor on the added ones.

The standard error grows when the added predictors are correlated with X_j (body fat, elections); when they are not, it barely changes.

The sample distribution with two predictors

Model with two standardized predictors with correlation $\rho_{1,2}$:

$$Y = \beta_1 X_1 + \beta_2 X_2 + \epsilon.$$

With one predictor, $\text{se}(\hat{\beta}_1) = \sigma_\epsilon / (\sigma_X \sqrt{N})$. With two, we need the *joint* sample distribution of $(\hat{\beta}_1, \hat{\beta}_2)$.

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Extreme case, $X_1 \approx X_2$:

$$Y \approx (\beta_1 + \beta_2) X_1 + \epsilon.$$

The data pin down $\beta_1 + \beta_2$, but not β_1 and β_2 separately: raising $\hat{\beta}_1$ and lowering $\hat{\beta}_2$ by the same amount fits almost equally well.

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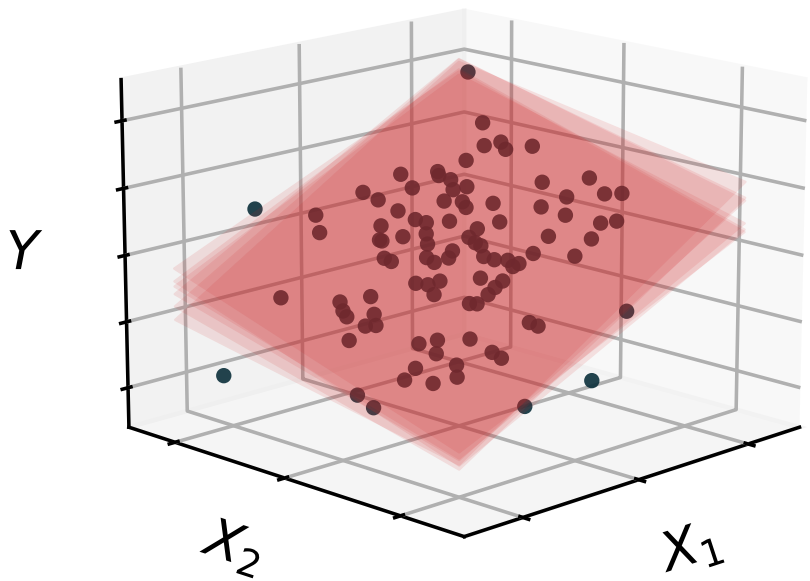
Question

If a new dataset from the same model gives a larger $\hat{\beta}_1$, what do you expect for $\hat{\beta}_2$?

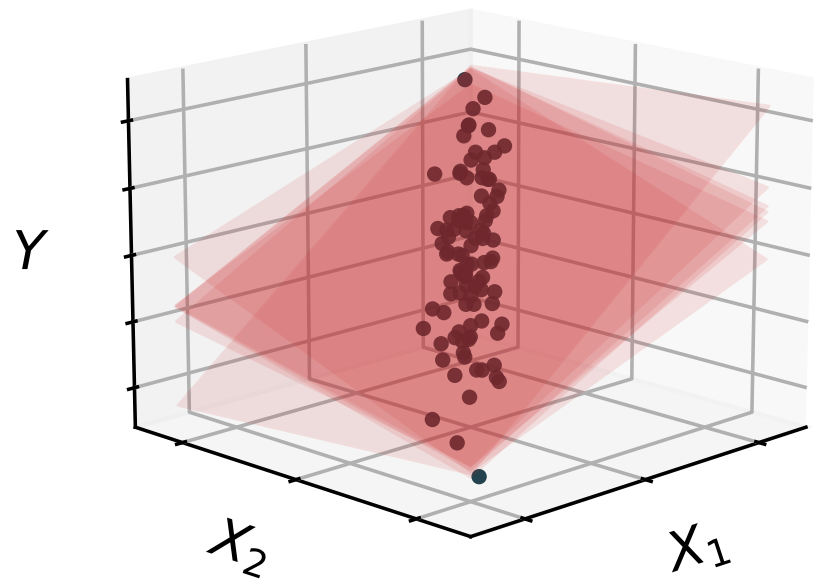
Why: the regression plane wobbles

Fitted planes from 6 datasets of the same model ($Y = X_1 + 0.5X_2 + \epsilon$, $N = 100$); points: one of the datasets.

$\rho_{1,2} = 0$



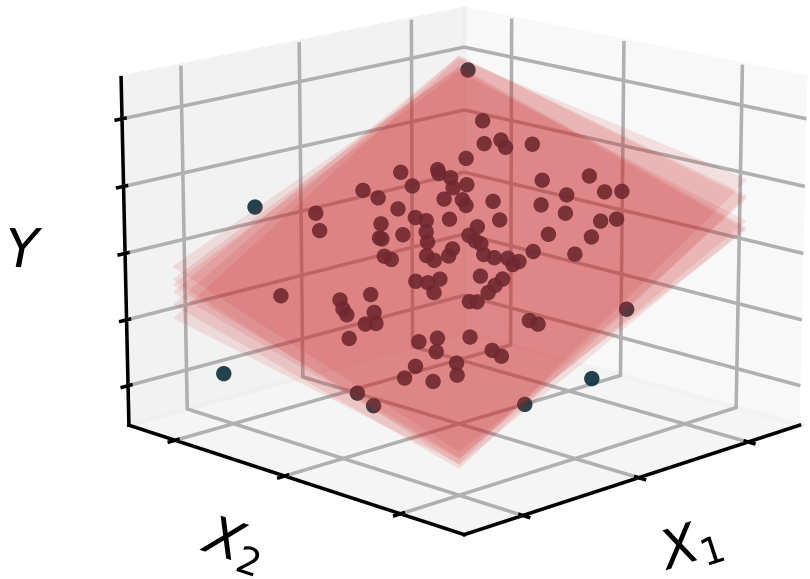
$\rho_{1,2} = 0.95$



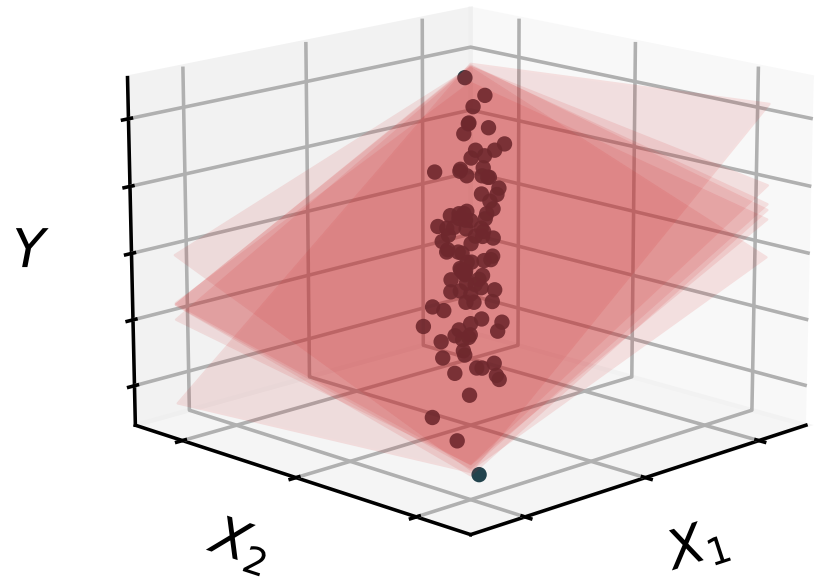
Why: the regression plane wobbles

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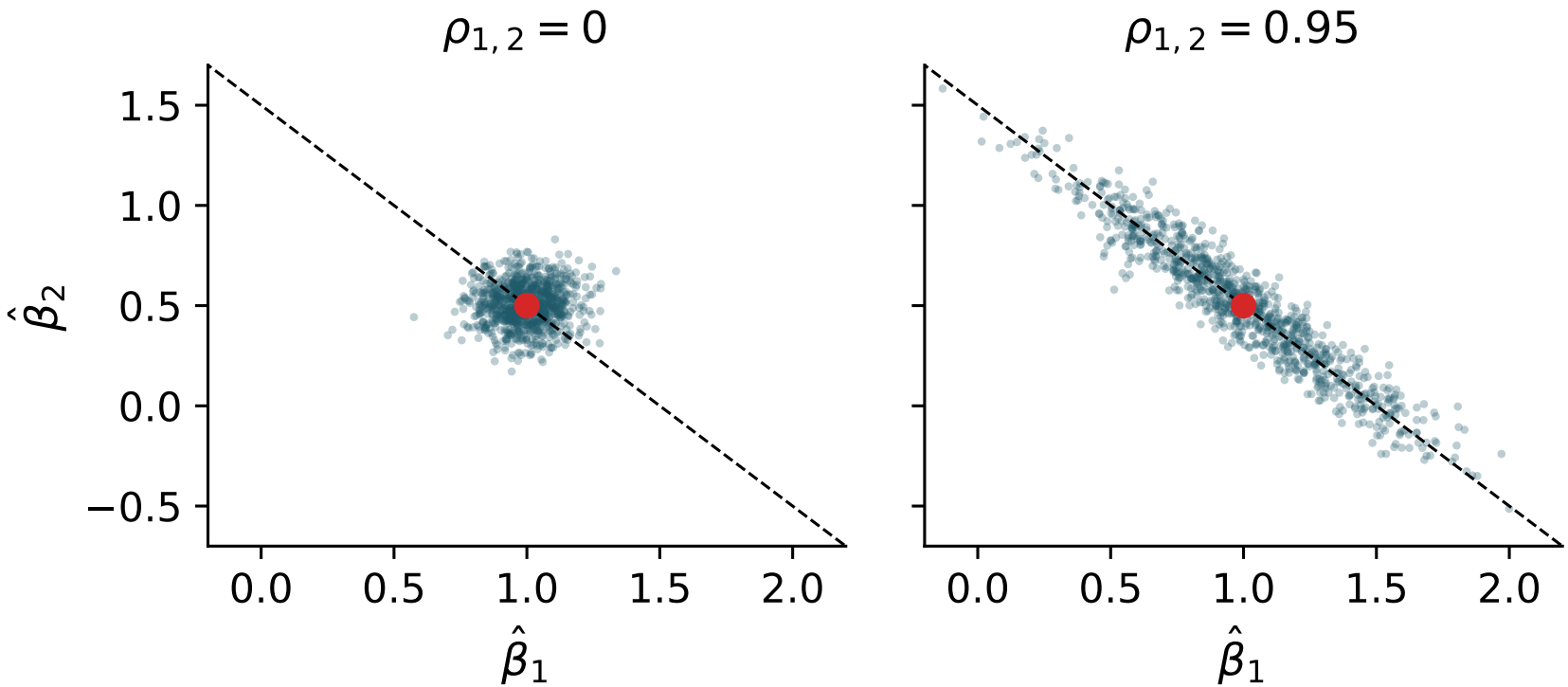
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With $X_1 \approx X_2$ the points lie near a line. The plane can tilt about that line without changing the fit much, so the individual slopes are poorly determined.

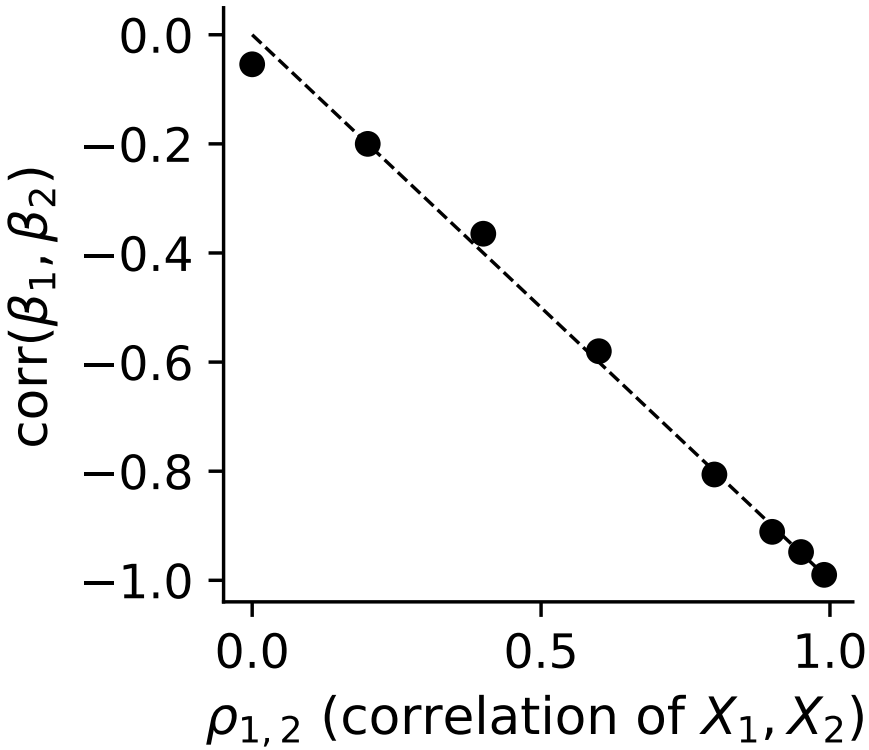
Simulated sample distribution

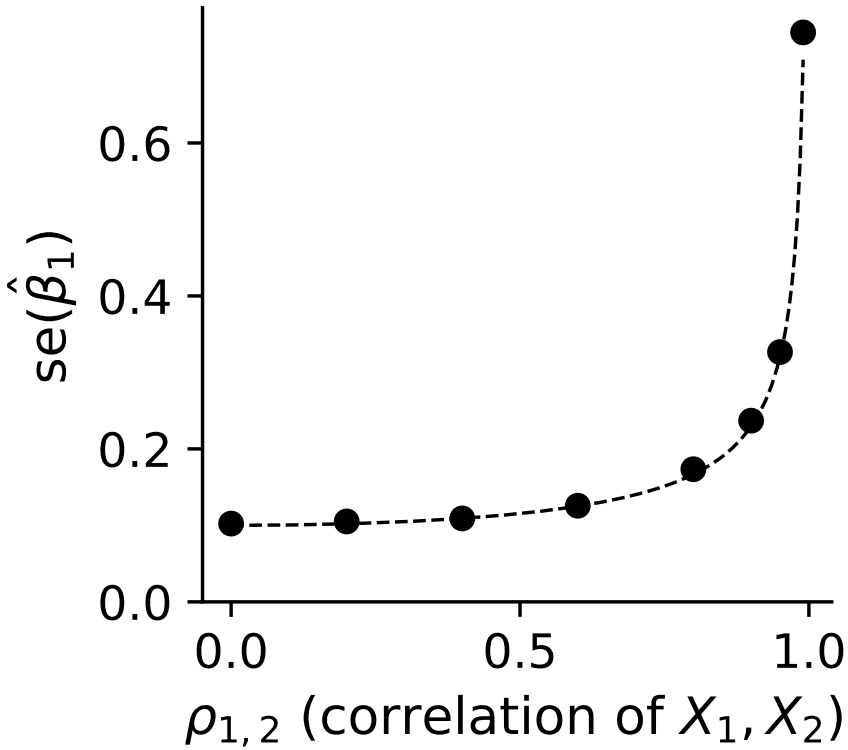
1000 datasets of $N = 100$ points from $Y = X_1 + 0.5X_2 + \epsilon$ ($\sigma_\epsilon = 1$):



Red: the true (β_1, β_2). Dashed: $\hat{\beta}_1 + \hat{\beta}_2 = 1.5$.

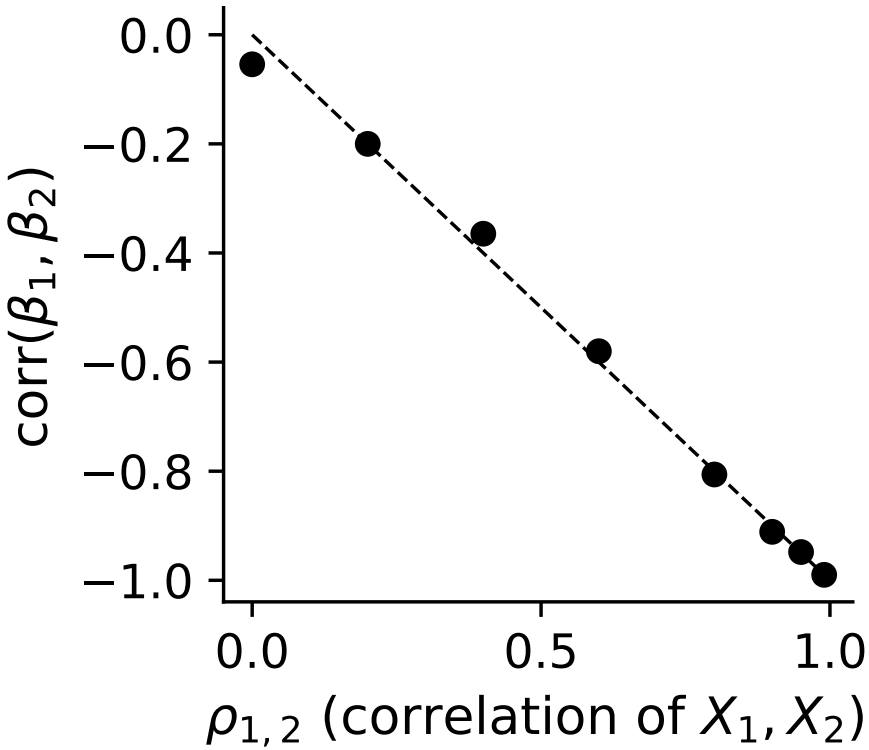
Correlation and standard error vs. $\rho_{1,2}$

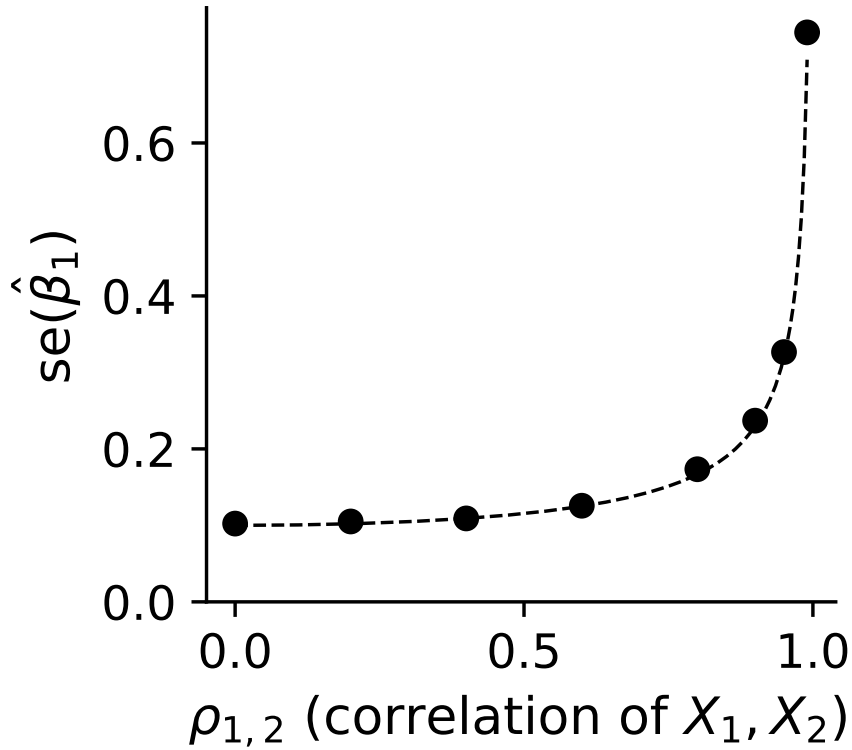




Points: 2000 simulated datasets for each $\rho_{1,2}$. Dashed: $\text{corr}(\hat{\beta}_1, \hat{\beta}_2) = -\rho_{1,2}$ and $\text{se}(\hat{\beta}_1) = \sigma_\epsilon / \sqrt{N(1-\rho_{1,2}^2)}$.

Correlation and standard error vs. $\rho_{1,2}$





Points: 2000 simulated datasets for each $\rho_{1,2}$. Dashed: $\text{corr}(\hat{\beta}_1, \hat{\beta}_2) = -\rho_{1,2}$ and $\text{se}(\hat{\beta}_1) = \sigma_\epsilon / \sqrt{N(1-\rho_{1,2}^2)}$.

At $\rho_{1,2} = 0.95$, $\text{se}(\hat{\beta}_1)$ is about 3 times larger than with uncorrelated predictors, but $\hat{\beta}_1 + \hat{\beta}_2$ is as precise as ever.

Real example: incumbency and past vote

In the election data, corr(`past_vote`, `inc`) $= 0.86$: incumbents mostly hold districts that already voted for their party.

	`inc` only	`past_vote` only	both
$\hat{\beta}_{\text{inc}}$ (se)	0.191 (0.004)		0.096 (0.007)
$\hat{\beta}_{\text{past}}$ (se)		0.918 (0.020)	0.521 (0.032)

In the model with both, the estimated corr$(\hat{\beta}_{\text{past}}, \hat{\beta}_{\text{inc}}) = -0.86$.

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Question

Why are both standard errors larger in the model with both predictors? Why would the incumbency advantage be hard to estimate if almost every incumbent held a safe seat?